Hu Zhiheng Henrey

Game Programmer

+65 9828 4968 | henrey.hu@gmail.com | www.linkedin.com/in/henrey-hu | github.com/HenreyHu/Portfolio

SUMMARY

Computer Science student at SIT and DigiPen Institute of Technology Singapore, specialising in Interactive Media and Game Development, seeking a 12 month game programming internship. Skilled in C++, C# and Unity, with hands-on experience in enemy AI behaviour, gameplay systems, UI frameworks and save systems. Completed 2 team games of 5 to 8 members, each built from scratch in a custom C++ engine, and now building a third. Prototyped enemy behaviour in Unity before building it in the engine, and integrated teammates' modules into one playable game. Brings teamwork and calm decision making under pressure from SCDF service and private tutoring. Aspires to grow as a gameplay programmer who builds responsive, polished games.

EDUCATION

Singapore Institute of Technology (SIT) & DigiPen Institute of Technology Singapore Aug 2024 - Present

Bachelor of Science (Honours) in Computer Science in Interactive Media and Game Development Expected Graduation Apr 2028

Relevant Modules Artificial Intelligence for Games, Modern C++ Design Patterns, Data Structures, Developing Immersive Applications, Linear Algebra

TECHNICAL SKILLS

Programming Languages	C++, C#, Python, JavaScript, SQL, HTML
Game Engines	Unity, Unreal Engine, Custom C++ engines
Operating Systems	Windows
Tools and Software	Git, Figma, Excel, Affinity, DaVinci Resolve

PROJECTS

Programmer | STRETCH, 2D platformer, 8-person team | C++, custom engine, RapidJSON, Unity C# (Sep 2025 - Apr 2026)

- Enemy behaviour needed to be proven before committing to it. Prototyped 4 behaviour types in Unity C#, then implemented them as a 4 state machine in C++. Owned enemy logic and UI as AI & UI Champion across 6 milestones.
- The game needed menus usable on keyboard, mouse and gamepad. Developed a UI framework of 10 screens with buttons and sliders, so one codebase supports all three input types.
- The team needed progress and level data to survive crashes and restarts. Built a JSON save and load system with RapidJSON covering about 30 data types, plus auto-save and categorised crash logging. Players resume from the last saved state and failures are traceable.
- Level designers needed to reverse mistakes in the team's in-house design tool. Solely designed and wrote an undo/redo system using JSON snapshots for edits, creation and deletion.

Programmer | WAVES, 2D stealth maze game, 5-person team | C++, custom engine (Jan 2025 - Apr 2025)

- Five teammates' screens had to run as one application. Designed a state management system where every screen follows one lifecycle, so screens switch cleanly with no memory leaks.
- Players needed clear navigation. Built the UI flow across 8 screens and overlays, with a reusable component library and font cache shared by all screens.
- Audio needed to be adjustable in game. Built a draggable volume slider linked to the global audio settings, and the rendering of walls and boundaries.
- Led integration of four teammates' modules into one working application with end conditions.

WORK EXPERIENCE

Singapore Civil Defence Force (SCDF) | Emergency Medical Technician Apr 2022 - Feb 2024

- Responded to high-stress emergency callouts on an ambulance crew over 22 months, making real-time triage and treatment decisions.
- Worked within a rotating team, coordinating roles and communicating clearly during each callout.
- Stayed calm and reliable under repeated pressure.

Private Tutor | Geography, Social Studies, Maths and Science 2020 - Present

- Tutored 4 students since 2020 in Geography, Social Studies, Maths and Science, building their confidence in problem solving.
- Adapted lessons to each student's pace and explained difficult concepts in simple terms.
- Raised one student from a fail to B3 in Geography and Social Studies.

CERTIFICATIONS

Google Data Analytics Professional Certificate (Coursera) Jan 2026 - Oct 2026

INTERESTS

Video/Photo Editing (Affinity, DaVinci Resolve), Dizi (Grade 9), Floorball (SCDF Inter-Division Champion, 2023)